CHE 312 HW #7

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#1 a) $q = m_c C_{pc} \Delta T_c = m_h C_{ph} \Delta T_h$

$q = (5{,}000)(210-90) = 600{,}000 \ BTU/h$

$600{,}000 = (10{,}000)(T_{1,h} - T_{2,h})$

$T_{2,h} = 200°F$

$$\Delta T_m = \frac{(200-90)-(260-210)}{\ln\left(\frac{200-90}{260-210}\right)} = 76.097°F$$

$$q = U_o A \Delta T_m \longrightarrow A = \frac{q}{U_o \Delta T_m} = \frac{600000}{(150)(76.097)} = 52.6 ft^2$$

$$L = \frac{A}{2\pi r_o N_t} = \frac{52.6}{\pi(0.5/12)(30)} \longrightarrow \boxed{L = 13.4 ft}$$

b) You can increase the number of tubes.

#2 a) $$t = -\frac{\rho C_p}{h A/V} \ln\left(\frac{T-T_a}{T_o-T_a}\right) = -\frac{900}{(500)(3/2700 \cdot 0.01)} \ln\left(\frac{40-20}{90-20}\right)$$

$\boxed{t = 20.3 \ s}$

b) $$M_s C_{ps} \frac{dT_s}{dt} = -hA(T_s - T_w), \quad M_w C_{pw} \frac{dT_w}{dt} = hA(T_s - T_w)$$

$$\frac{d\Delta T}{dt} = -hA\left(\frac{1}{M_s C_{ps}} + \frac{1}{M_c C_{pc}}\right)\Delta T$$

$$K' = hA\left(\frac{1}{M_s C_{ps}} + \frac{1}{M_c C_{pc}}\right) \Rightarrow \Delta T(t) = \Delta T_0 e^{-K't}$$

$$K' = (500)\left(\frac{3}{2700 \cdot 0.01}\right)\left(\frac{1}{900} + \frac{1}{4186}\right) = 0.075 \ s^{-1}$$

$$T_{eq} = \frac{M_s C_{ps} T_{0,s} + M_w C_{pw} T_{0,w}}{M_s C_{ps} + M_w C_{pw}} = \frac{(900)(90) + (4186)(20)}{900 + 4186} = 32.39°C$$

$$T_s(t) = T_{eq} + \frac{M_w C_{pw}}{\text{[illegible]}} \Delta T_0 e^{-K't} = 32.39 + \frac{4186}{\text{[illegible]}}(70) e^{-0.075t}$$

$$T_s(t) = T_{eq} + \frac{M_w C_{pw}}{M_s C_{ps} + M_w C_{pw}} \Delta T_0 e^{-K't} = 32.39 + \frac{4186}{41860+900}(70)e^{-0.075t}$$

$$40 = 32.39 + \frac{4186}{4186+900}(70)e^{-0.075t} \longrightarrow \boxed{t = 27.0s}$$